

Esempi di applicazioni delle serie di Fourier.

① ODE : Ricerca di soluzioni periodiche di eq. lineari

$$(*) \quad \sum_{j=0}^n a_j u^{(j)} = f \quad x \in \mathbb{R}$$

$\uparrow$ coeff costanti $\uparrow$ funzione in $L^2_T(\mathbb{R})$

Domanda: esistono soluzioni T -periodiche? (in $L^2_T(\mathbb{R})$).

$$P(\lambda) = \sum_{j=0}^n a_j \lambda^j \quad \text{polinomio caratteristico.}$$

Esempio

$$\left[u'' + u = \sin(\omega x) \quad \omega \in \mathbb{R}^+ \quad T = \frac{2\pi}{\omega} \right]$$

$$\sum_{j=0}^n a_j \widehat{u^{(j)}}_k = \widehat{f}_k \quad \forall k \in \mathbb{Z}$$

$$(**) \quad \left[\sum_{j=0}^n a_j (i\xi_k)^j \right] \widehat{u}_k = \widehat{f}_k \quad \forall k \in \mathbb{Z}$$

$$\boxed{1 \text{ ODE in } u \rightsquigarrow \infty \text{ eq. ALGEBRICHE in } \widehat{u}_k}$$

$$(**) \quad \underset{\substack{\uparrow \\ \text{pol. caratteristico}}}{P(i\xi_k)} \widehat{u}_k = \widehat{f}_k \quad \forall k \in \mathbb{Z}.$$

$\swarrow$ dato $\searrow$ incognita

$$a x = b$$

$$\nearrow a \neq 0 \Rightarrow \text{unique solution } x = \frac{b}{a}$$

$$\searrow a = 0 \Rightarrow \begin{cases} b = 0 & \text{infinite solutions} \\ b \neq 0 & \text{no solutions} \end{cases}$$

Caso 1

$$P(i\xi_k) \neq 0 \quad \forall k \in \mathbb{Z} \Rightarrow$$

$$\hat{u}_k = \frac{\hat{f}_k}{P(i\xi_k)}$$

$$\text{unique solution } u \in L_T^2(\mathbb{R})$$

$$u = \sum_{k \in \mathbb{Z}} \hat{u}_k e^{i\xi_k x}$$

Caso 2

$$P(i\xi_k) = 0 \quad \forall k \in \{K_1^*, \dots, K_p^*\}$$

$$\nearrow \hat{f}_k = 0 \quad \text{per } k = \underset{T}{K_1^*}, \dots, \underset{T}{K_p^*} \Rightarrow \text{infinite solutions in } L_T^2(\mathbb{R})$$

$$\hat{u}_k = \begin{cases} \frac{\hat{f}_k}{P(i\xi_k)} & k \neq K_1^*, \dots, K_p^* \\ \text{ARBITRARI} & k = K_1^*, \dots, K_p^* \end{cases}$$

$$\searrow \hat{f}_k \neq 0 \quad \text{per } k \in \{K_1^*, \dots, K_p^*\} \Rightarrow \text{No solutions in } L_T^2(\mathbb{R})$$

Esempio:

$$u'' + u = \sin(\omega x) \quad \omega \in \mathbb{R}^+ \quad T = \frac{2\pi}{\omega}$$

$$\circ \quad \xi_k = k \cdot \frac{2\pi}{T} = k\omega$$

$$\circ \quad P(\lambda) = \lambda^2 + 1$$

Lemma: $P(i k \omega) \neq 0 \quad \forall k \in \mathbb{Z}$.

$$(i k \omega)^2 + 1 = -k^2 \omega^2 + 1 \iff$$

$$\omega^2 \neq \frac{1}{k^2} \iff \omega \neq \frac{1}{k} \quad k \in \mathbb{N} \setminus \{0\}$$

$$\hat{u}_k = \frac{\hat{f}_k}{P(i k \omega)} = \frac{\hat{f}_k}{P(i k \omega)} = \frac{\hat{f}_k}{1 - k^2 \omega^2}$$

$$f = \sin(\omega x) = \frac{e^{i\omega x} - e^{-i\omega x}}{2i} \Rightarrow \hat{f}_k = \begin{cases} 0 & (k \neq \pm 1) \\ \pm \frac{1}{2i} & (k = \pm 1) \end{cases}$$

$$\hat{u}_k = \begin{cases} 0 & k \neq \pm 1 \\ \pm \frac{1}{2i(1 - \omega^2)} & k = \pm 1 \end{cases}$$

$$u = \frac{1}{1 - \omega^2} \cdot \underbrace{\frac{1}{2i} (e^{i\omega x} - e^{-i\omega x})}_{\sin(\omega x)} = \frac{1}{(1 - \omega^2)} \sin(\omega x)$$

Caso 2

$$\omega = \frac{1}{k^*} \quad k^* \in \mathbb{N} \setminus \{0\}$$

→ $\hat{f}_{k^*} = 0$

∞ soluzioni T -periodiche

$$\omega = \frac{1}{k^*} \neq 1$$

$$\hat{u}_k = \begin{cases} \frac{\pm 1}{2i} \cdot \frac{1}{1-\omega^2} & k = \pm 1 \\ 0 & \text{altrimenti} \\ \text{arbitrari} & k = \pm k^* \end{cases}$$

$$\begin{aligned} u(x) &= \frac{\sin(\omega x)}{1-\omega^2} + \underbrace{c_1}_{\in \mathbb{R}} e^{i\omega k^* x} + \underbrace{c_2}_{\in \mathbb{R}} e^{-i\omega k^* x} \\ &= \frac{\sin(\omega x)}{1-\omega^2} + c_1 e^{ix} + c_2 e^{-ix} \quad c_1, c_2 \in \mathbb{R} \end{aligned}$$

→ $\hat{f}_{k^*} \neq 0$

non ci sono soluzioni in $L^2_T(\mathbb{R})$.

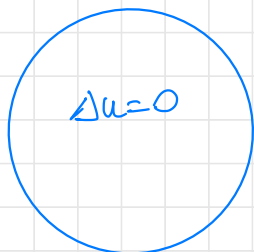
$$\omega = \frac{1}{k^*} = 1$$

2) PDE Pb. di Dirichlet e di Neumann
per l'eq. di Laplace nel disco

$$\Omega = B_R(0) = \{x \in \mathbb{R}^2 : x^2 + y^2 < R\}$$

$$(*) \quad \Delta u = 0 \quad \text{in } \Omega$$

$$u_{xx} + u_{yy}$$



$$(D) \quad \begin{cases} \Delta u = 0 & \text{in } B_R(0) \\ u = g & \text{su } \partial B_R(0) \end{cases}$$

$$(N) \quad \begin{cases} \Delta u = 0 & \text{in } B_R(0) \\ \frac{\partial u}{\partial \nu} = g & \text{su } \partial B_R(0) \end{cases}$$

$$u_{\#}(\rho, \theta) = u(\rho \cos \theta, \rho \sin \theta) \quad \begin{matrix} \rho \in [0, R) \\ \theta \in \mathbb{R} \end{matrix}$$

$$\theta \mapsto u_{\#}(\rho, \theta) \quad 2\pi\text{-periodica}$$

$$u_{\#}(\rho, \cdot) \in L^2_{2\pi}(\mathbb{R})$$

$$\Delta u = \frac{\partial^2 u_{\#}}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial u_{\#}}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 u_{\#}}{\partial \theta^2}$$

LAPLACIANO DI COORD. POLARI

$$\forall \rho \neq 0$$

$$u_{\#}(\rho, \theta) = \sum_k \underbrace{\hat{u}_k(\rho)}_{\uparrow} e^{ik\theta} \quad \forall \rho \neq 0$$

Supponiamo di poter derivare per serie

$$\sum_k \left[\left(\frac{d^2}{d\rho^2} \hat{u}_k(\rho) \right) e^{ik\theta} + \left(\frac{1}{\rho} \frac{d}{d\rho} \hat{u}_k(\rho) \right) e^{ik\theta} + \frac{1}{\rho^2} \hat{u}_k(\rho) (-k^2) e^{ik\theta} \right] = 0$$

$$\left(\sum_k \right) \left[\frac{d^2}{d\rho^2} \hat{u}_k(\rho) + \frac{1}{\rho} \frac{d}{d\rho} \hat{u}_k(\rho) - \frac{k^2}{\rho^2} \hat{u}_k(\rho) \right] e^{ik\theta} = 0$$

(*) PDE in $u \rightsquigarrow$

$$(**) \quad \frac{d^2}{d\rho^2} \hat{u}_k(\rho) + \frac{1}{\rho} \frac{d}{d\rho} \hat{u}_k(\rho) - \frac{k^2}{\rho^2} \hat{u}_k(\rho) = 0 \quad \forall k \in \mathbb{Z}$$

sistema di infinite ODE.

$$\rho^2 \frac{d^2}{d\rho^2} \hat{u}_k(\rho) + \rho \frac{d}{d\rho} \hat{u}_k(\rho) - k^2 \hat{u}_k(\rho) = 0$$

eq. di tipo "Eulero".

Chiamiamo soluzioni di tipo $\hat{u}_k(\rho) = \rho^\alpha$

$$\frac{d}{d\rho} \hat{u}_k(\rho) = \alpha \cdot \rho^{\alpha-1} \quad \frac{d^2}{d\rho^2} \hat{u}_k(\rho) = \alpha(\alpha-1) \rho^{\alpha-2}$$

Inserisco.

$$\alpha(\alpha-1) \rho^\alpha + \alpha \rho^\alpha - k^2 \rho^\alpha = 0$$

$$\rho^\alpha \left[\underbrace{\alpha(\alpha-1) + \alpha - k^2}_0 \right] = 0$$

$$\alpha^2 - \alpha + \alpha - k^2 = 0$$

$$\alpha^2 = k^2 \quad \alpha = \pm |k|$$

$\rightarrow k \neq 0$ $\hat{u}_k(\rho) = c_k \rho^{|k|} + c'_k \rho^{-|k|}$

$\rightarrow k=0$ $\hat{u}_k(\rho) = c_0 + c'_0 \log \rho$

$$\rho^2 \frac{d^2 \hat{u}_0}{d\rho^2} + \rho \frac{d\hat{u}_0}{d\rho} = 0 \quad \hat{v}_0 = \frac{d\hat{u}_0}{d\rho}$$

$$\rho \frac{d\hat{v}_0}{d\rho} + \rho v_0 = 0$$

$$\frac{\frac{d\hat{v}_0}{d\rho}}{\hat{v}_0} = -\frac{1}{\rho} \Rightarrow \hat{v}_0(\rho) = \frac{c'_0}{\rho}$$

$$\Rightarrow \hat{u}_0(\rho) = c'_0 \log \rho$$

$$u_{\#}(p, \theta) = c_0 + c'_0 \log p + \sum_{k \neq 0} [c_k p^{|k|} + c_k p^{-|k|}] e^{ik\theta}$$

$$u_{\#}(p, \theta) = c_0 + \sum_{k \neq 0} c_k p^{|k|} e^{ik\theta}$$

$$u_{\#}(p, \theta) = \sum_{k \in \mathbb{Z}} c_k p^{|k|} e^{ik\theta}$$

↑
generica funzione armonica in $B_R(0)$

purché si potesse derivare per serie!